Chapter 43 - Electric Circuits
Calculus Physics II

Circuit – is any conducting path that charge can flow. More specifically, it is the total electrical path made from conducting wire, power sources and circuit elements.

In order to have a functioning and useful electric device there should be

1) Power source – Source of the electrical field that moves the charge around the circuit such as a battery, capacitor, power supply etc. … The power source is the source of the energy given to the charge by way of the electric field.

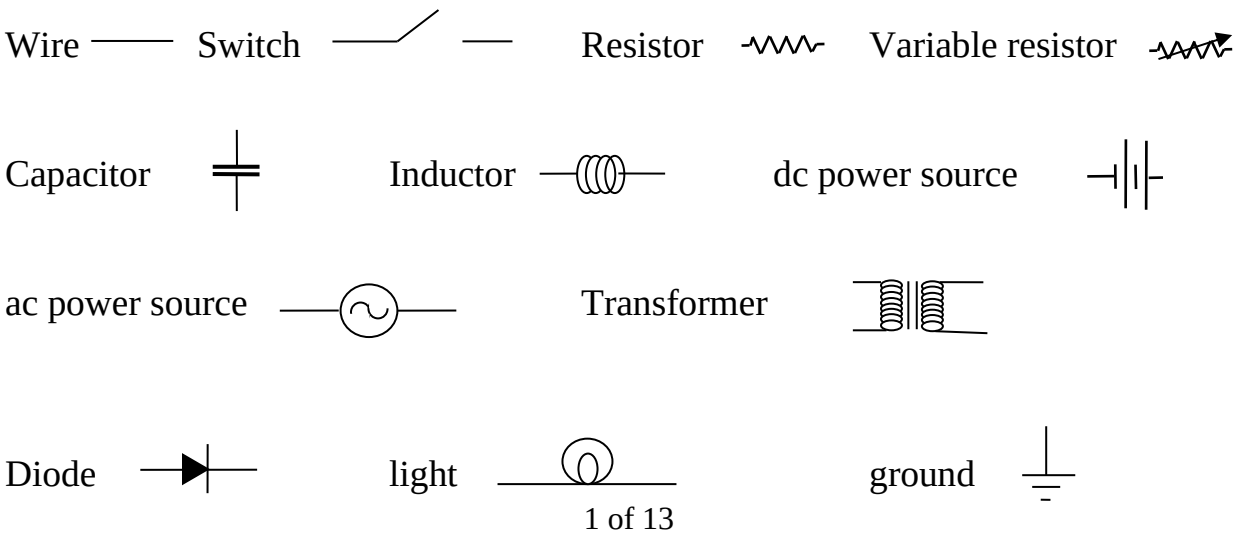
2) Circuit – A conducting path for the charge to flow. This is most commonly accomplished by using copper wire but not always. Any conducing material can be used as a circuit.

3) Load - all the devices, circuit elements or components in the circuit that use the energy from the electric field. The load does not supply any of the energy to the circuit but is powered by the circuit.

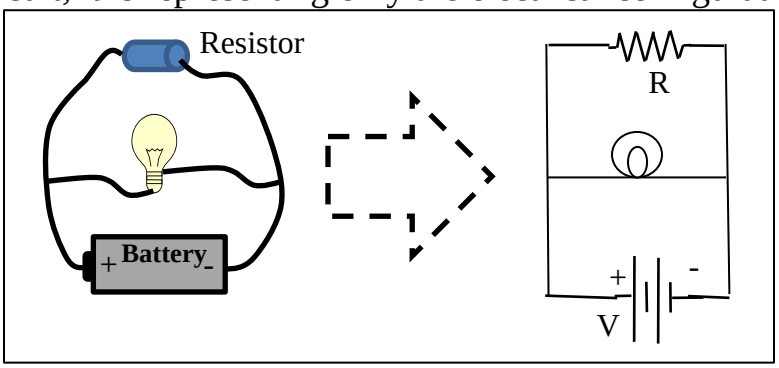
4) Current and voltage sensors – it is often useful to monitor the current or voltage in a device to determine how it is changing with changing conditions. This can make the electric device any type of sensor by how the voltage and current change with what is being sensed. Example: multimeters, thermometer, odometer, heart monitor, motion sensor, phone, etc. …

Circuit Diagrams – are representations of an actual circuit using symbols and lines to show the circuit elements and how they are wired. It is a graphical representation of the electric circuit. It is also known as an electrical schematic.

Electric Circuit Symbols



Example of a circuit diagram: The circuit below shows a battery hooked up to a light bulb and resistor. This circuit can be represented by using the symbols for each device and connecting them by lines (wire) as they are in the actual circuit. It is only a schematic and does not have to look like the actual circuit, it is representing only the electrical configuration.

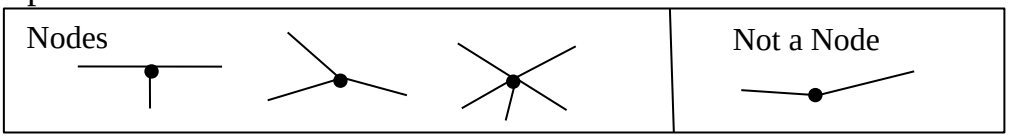


Note: For the dc power source symbol, the long line indicates the positive side, and the short side is the negative side.

Circuits Configurations – Circuit elements can be "wired" in many different configurations, but Ohm's law can only be applied to a single power source, resistance and current. In order to solve more complicated systems an equivalent resistance must be determined that is equal to the total resistance that the power source experiences due to all the circuit elements. Once an equivalent resistance is determined then Ohm's law can be applied using the equivalent resistance.

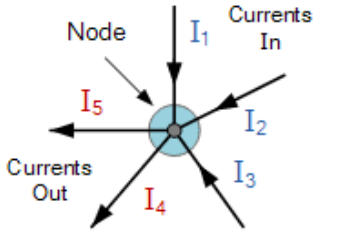
Equivalent Resistant - is a single resistance that is equal to the entire circuit's resistance being careful to take all types of configurations (series and parallel) of the circuit elements.

Nodes or Junction – A node is a point where three or more current-carrying wires or branches meet. The current will split at the node in different paths (wires) with different values of the current that will depend on the resistance in each branch.



✓ All currents going into a node must be equal to all currents going out of a node.

$$\Sigma I_{in} = \Sigma I_{out}$$



$I_1 + I_2 + I_3 = I_4 + I_5$

Current Entering the Node

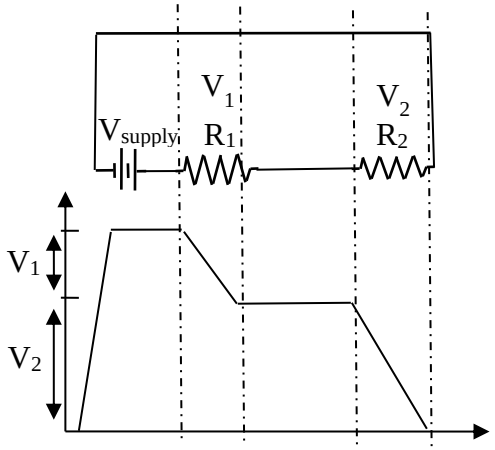
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Current Leaving the Node

Resistors in Series - A Resistor is in series with other resistors when current must flow through each resistor one after the other in sequence. The output of one resistor is the input for the next resistor until they connect back to the power source. The series circuit has no nodes.

- In series circuits the current (I) remains constant.
- In series circuits the voltage drops across each resistor adds up to the total voltage supplied by the power source.

For a circuit with resistors in series



Current through each resistor

$$I_T = I_1 = I_2 = I_3 = \ldots = I_{supply}$$

Voltage across each resistor

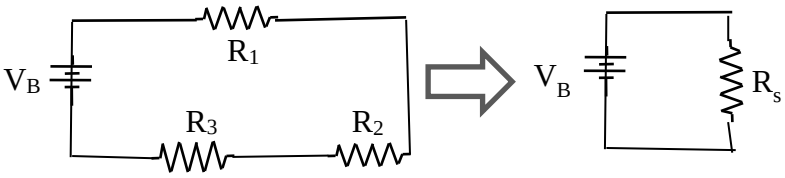
$$V_T = V_1 + V_2 + V_3 + \ldots = V_{supply}$$

$$V_T = IR_{eq} = IR_1 + IR_2 + IR_3 + \ldots$$

The graph shows the voltage that is across each resistor (R_1 and R_2) and the total voltage across the circuit $V_{supply} = V_1 + V_2$.

There is no voltage drop across the wires, they are assumed to have no resistance.

Series Equivalent Resistance's (R_{eq} or R_s) - Resistors wired in series, the current must flow through each resistor in sequence, each resistor that is added in series adds to the total resistance of the circuit. The equivalent resistance for resistors wired in series is the sum of all resistors.



The power dissipated by each resistor is equal to the total power used by the circuit.

$$P_{Total} = P_1 + P_2 + P_3 + \cdots = IR_{total} = IR_1 + IR_2 + IR_3 + \cdots$$

Knowing that the current in a series circuit is constant

$$R_{total} = R_1 + R_2 + R_3 + \cdots$$

The equivalent resistance R_{eq} wired in series is equal to the sum of the individual resistors and is given by

$$\boxed{R_s = R_1 + R_2 + R_3 + \ldots = R_{eq} \quad \text{For series circuit}}$$

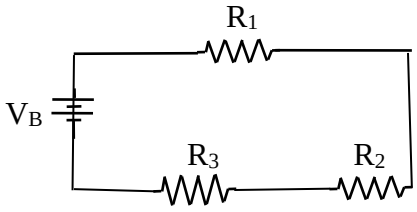
$$V = IR_s \quad \text{or} \quad I = V/R_s$$

For the above circuit the voltage drop across each resistor is given by

$$V_1 = IR_1\,, \quad V_2 = IR_2 \quad \text{and} \quad V_{supply} = V_1 + V_2$$

Example: Three resistors of varying resistance are wired in series as shown below.

$R_1 = 10\ \Omega$
$R_2 = 15\ \Omega$
$R_3 = 20\ \Omega$
$V_B = 24\ V$



a) Determine the equivalent resistance in the circuit?

$R_s = R_1 + R_2 + R_3 = 10\ \Omega + 15\ \Omega + 20\ \Omega = 45\ \Omega$

b) Determine the current flowing through the 6 Ω resistor?

$V = IR_s$ Therefore

$I = V/R_s = 24V/45\Omega = 0.53\ A$

c) Determine the voltage drop across each resistor?

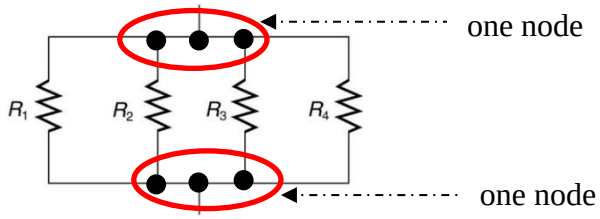
$V_1 = IR_1 = (0.53A)(\ 10\Omega) = 5.33\ V$
$V_2 = IR_2 = (0.53A)(15\Omega) = 7.995\ V$
$V_3 = IR_3 = (0.53A)(20\Omega) = 10.66\ V$

❖ Note that the larger the resistance the larger the voltage drop across the resistor.

$V_{supply} = V_1 + V_2 = 5.33\ V + 7.995\ V + 10.66\ V = 24\ V$

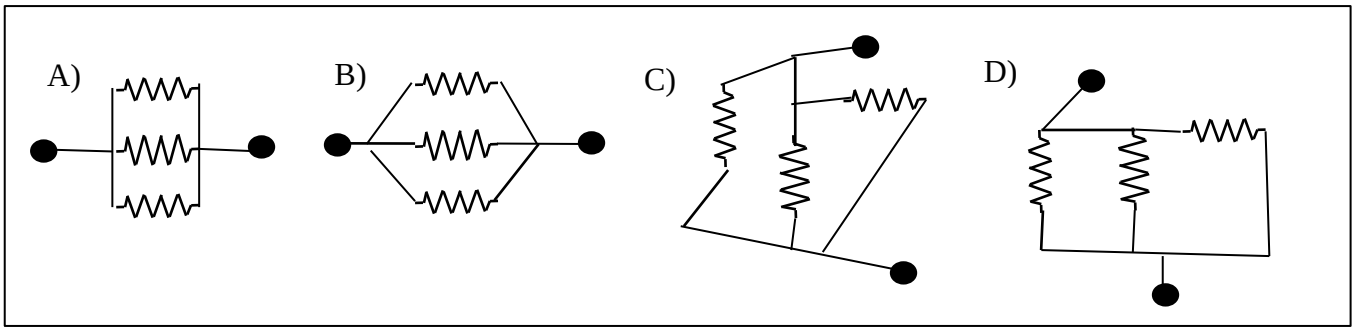
The sum of the voltage drops across the series resistor is equal to the voltage of the power supply.

Resistors in Parallel – When two or more resistors are connected in parallel the ends of the resistors are connected to the same node or nodes that are connected by a wire only. The same side of all the resistors are connected to the same node and all of the opposite side of the resistors are connected to a different node.



- When wires are connected with NO resistors (or any other circuit element) between the nodes then all the wires are the SAME node!

The diagram below shows three resistors wired in parallel, the black dots are the start and ending of the circuit. Note that all 4 diagrams represent the exact same three resistors wired in parallel, and all 4 diagrams would have the same equivalent resistance.



For the circuits above

A) All the wires on the left side are the same node therefore all the wires on the left are connected to the same node. The same goes for the right side.
B) All the resistors are wired to the same node on the left and to the second node on the right.
C) If wires are followed on the top, all the wires are connected and there are no other elements therefore they are the same node. Following the wires on the bottom the same can be seen, it is the second node.
D) It can be seen that the first two resistors are in parallel, the third resistor is wired on the top to the first node and on the bottom to the second node therefore it is in parallel with the other two resistors.

- In parallel circuits the voltage drop across each resistor is the same and equal to the voltage drop across the nodes.
- In parallel circuits the current going into each node is equal to the current going out of each node.

For a Parallel circuit

Current through each resistor in a parallel circuit

$I_T = I_1 + I_2 + I_3 + \ldots = I_{supply}$

Voltage across each resistor in a parallel circuit

$V_1 = V_2 = \ldots = V_{node}$

Parallel Equivalent Resistant's (R_{eq} or R_p) - In a Parallel circuit the voltage (V) remains constant across the resistors in parallel and the potential drop across the resistors is equal to the voltage drop across the nodes ($V_T = V_1 = V_2 = V_3 = \ldots$) but the current splits at the nodes so $I_T = I_1 + I_2 + I_3 + \ldots$.

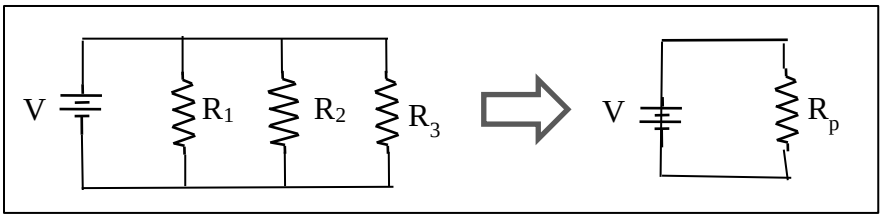
The power dissipated by each resistor is equal to the total power used by the circuit.

$$P_{Total} = P_1 + P_2 + P_3 + \cdots = \frac{V^2}{R_{total}} = \frac{V^2}{R_1} + \frac{V^2}{R_2} + \frac{V^2}{R_3} + \cdots$$

Knowing that the voltage across each resistor is the same value

$$\frac{1}{R_{total}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots$$

Therefore, the equivalent resistance is given by

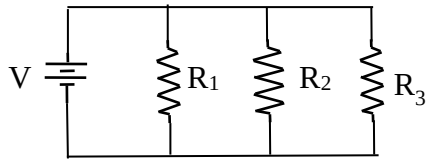


$$\boxed{\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots}$$

$V = IR_{eq}$ or $I = V / R_{eq}$

$$R_p = \frac{1}{\left[\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots\right]}$$

Example: Three resistors are wired in parallel as shown, if $R_1 = 10\ \Omega$, $R_2 = 15\ \Omega$, $R_3 = 20\ \Omega$ and V = 24 V



a) What is the equivalent resistance of the circuit?

$$R_p = \frac{1}{\left[\frac{1}{10\Omega} + \frac{1}{15\Omega} + \frac{1}{20\Omega}\right]} = 4.614\ \Omega$$

b) What current does the battery supply to the circuit?

$V = IR_p$ therefore $I = V/R_p = 5.2$ A

c) What is the current through each resistor? Because the voltage drop across the resistor in parallel is the same for all three resistors then

$I_1 = V/R_1 = 24V/10\Omega = 2.4$ A
$I_2 = V/R_2 = 24V/15\Omega = 1.6$ A
$I_3 = V/R_3 = 24V/20\Omega = 1.2$ A

Therefore, the total current is equal to the sum of the currents in each of the branches.

$I_{total} = I_1 + I_2 + I_3 = 2.4$ A + 1.6 A + 1.2 A = 5.2 A (same as part b)

Comparing Series and Parallel circuits

Series Circuits

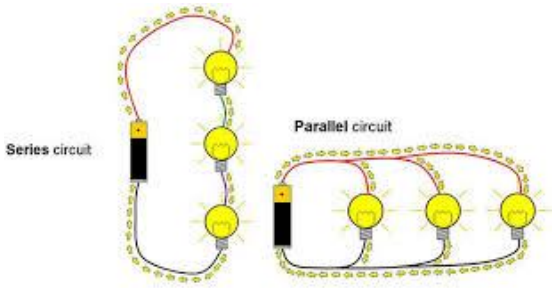
Current through each resistor: $I_T = I_1 = I_2 = I_3 = \ldots = I_{supply}$
Voltage across each resistor: $V_T = V_1 + V_2 + V_3 + \ldots = V_{supply}$
Equivalent resistance Series: $R_s = R_1 + R_2 + R_3 + \ldots$

Parallel Circuits

Current through the circuit $I_T = I_1 + I_2 + I_3 + \ldots = I_{supply}$
Voltage across each resistor: $V_1 = V_2 = \ldots = V_{node}$
Equivlent Resistance Parallel $R_p = \frac{1}{\left[\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right]}$

Assume the same light bulbs are wired in series and in parallel

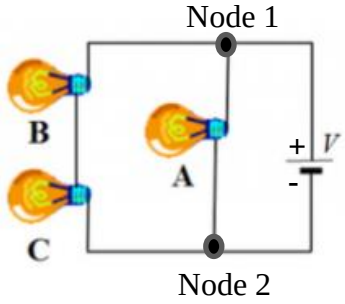
- The equivalent resistance in a series circuit is larger than in a parallel circuit given the same resistors, the light bulbs in series would have a large equivalent resistance.
- Because the resistance is larger in a series circuit as comparted to a parallel circuit the current flowing through a series circuit is less than a parallel circuit given the same supply voltage. The light bulbs in the parallel circuit would be brighter than the series bulbs.



- The more light bulbs that are added to the series circuit the dimmer the bulbs will become and the less current is supplied from the power supply. The more light bulbs that are added to a parallel circuit, the bulbs' brightness does not change but the more current the power supply must supply to the circuit.
- If a light bulb burns out (open circuit) in a series circuit, then all the light bulbs would go out but if a light bulb burns out in a parallel circuit, then only that bulb would be out and the rest would still be on.
- In order to have the series light bulbs the same brightness as the parallel light bulbs the potential in the series circuit must be increased.

Circuits wired Partially in Series and Partially in Parallel - To determine the equivalent resistance of a circuit with both series and parallel components, the circuit must be simplified by adding two or more resistors that are only in series or only in parallel, then simplify and recalculate using this new equivalent resistance, treating it as a single resistor, until a single equivalent resistance is obtained.

- The diagram shows three light bulbs wired in a circuit, bulbs B and C are wired in series and bulbs B and C are wired in parallel with bulb A.
- The current entering node #1 is the current supplied by the battery, it will split at the node but not evenly, (assuming the bulbs are identical)
- more current will flow through bulb A because of the smaller resistance. Therefore, bulb A will be brighter than bulbs B or C.
- The current flowing through bulbs B and C will be the same but dimmer than A.

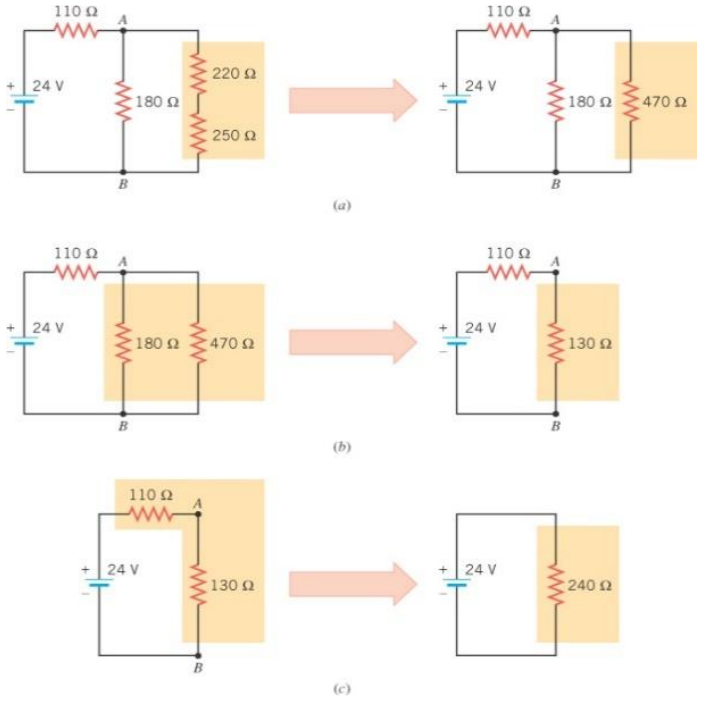


- At node # 2 the current from bulb A and from bulb B & C will recombine and go back into the power supply. The current out of the power supply is equal to the current back into the power supply.
- If bulb A is removed, bulb B & C will not change the brightness.
- If bulb B is removed, then bulb C will go out, but bulb A will not change its brightness.

Process for determining equivalent resistance for series/parallel circuit combination. To determine the final total equivalent resistance, identify the part of the circuit that can be simplified and redrawn with that equivalent resistance then identify and calculate what new equivalent resistances can be simplified. Keep repeating the above process until there is only one equivalent resistance.

I. Start with any resistors in series along a single wire with no branches between them and determine the equivalent resistance for resistor in series.
Or find two or more resistors that are parallel with each other with no other branches and only one resistor in each branch and find the equivalent resistant for resistors in parallel.
II. Draw new circuit with equivalent resistance in place of original resistors.
III. Repeat process until circuit has only one equivalent resistance.

Example: Using the diagram with 4 resistors in series and parallel below.



A) Determine the equivalent resistance of the circuit below?

1. From part a) the 220 and 250 resistors are in series therefore

$$R_{eq1} = 220\Omega + 250\Omega = 470\Omega$$

2. From part b) the R_{eq1} and the 180 Ω resistors are in parallel

$$R_{eq2} = \frac{1}{\frac{1}{470\Omega} + \frac{1}{180\Omega}} = 130\ \Omega$$

3. From part c) the R_{eq2} and the 110 Ω are in series

$$R_{eq3} = 130\Omega + 110\Omega = 240\Omega = R_{eq}$$

240 Ω is the equivalent resistance for the circuit

B) Determine the current supplied to the circuit by the battery?

$V = IR_{eq}$ therefore $I = V/R_{eq} = 0.1$ A

C) What is the current through the 110 Ω resistor? Because this resistor is between the power supply and the rest of the circuit and there are no junctions between it and the power supply the current through the 110 resistor is the same as from the power supply which is 0.1 A.

D) Determine the current through the 180 Ω resistor and the 220Ω & 250Ω resistors?

Looking at part b) the equivalent resistance for the 180 Ω and 470 Ω resistors was 130 Ω and this equivalent resistance was in series with the 110 Ω resistor, the same current must be going through both, therefore

$V_{180\Omega} = V_{130\Omega} = IR_{130\Omega} = (0.1A)(130\Omega) = 13$ V using this voltage

$I_{180\Omega} = V_{130\Omega}/R_{180\Omega} = (13V)/(180\Omega) = 0.0722$ A

Because the 180 Ω resistor is in parallel with the 220/250 Ω resistors the voltage drop across the two are the same but their equivalent resistance is 470 Ω therefore

$V_{220/250} = 13$ V

$I_{220/250} = V_{130\Omega}/R_{470\Omega} = (13V)/(470\Omega) = 0.0276$ A

$I = I_{180\Omega} + I_{220/250} = 0.0722$ A $+ 0.0276$ A $= 0.1$ A

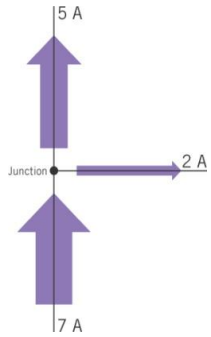
Kirchhoff's Rules - Kirchhoff's Rules are used when the simple application of Ohms Law cannot be applied then Kirchhoff's Rules must be used. Example: more than one power source.

Applying Kirchhoff's Rules involves using two fundamental principles, the Junction Rule and the Loop Rule, to analyze complex electrical circuits. They are especially useful for circuits that cannot be simplified using basic series and parallel resistance formulas.

Conservation of charge/current - Junction Rule – The sum of the magnitude of the currents directed into a junction or node equals the sum of the magnitudes of the currents directed out of the node; (Conservation of Charge)

$$\Sigma I_{in} = \Sigma I_{out}$$

All Current into the node = All current out of the node



- The total charge per second entering a node must equal the total charge per second leaving the node.

Loop rule (energy in = energy out) – Around any closed-circuit loop, the sum of the potential drops equals the sum of the potential rises.

$$\Sigma V_{Rises} = \Sigma V_{Drops}$$

Potential Drops = Potential Rises
Or

$$\Sigma V_{Rises} - \Sigma V_{Drops} = 0$$

- The sum of all the potential drops and rises equals zero, where the potential rises are positive values and the potential drops are negative values.

- The potential drops or rises depend on the direction of the current through the resistor.
- For a resistor the current direction is chosen, and the positive end of the resistor is where the current enters.
- The negative end for the resistor is where the current exits.

Potential Rise – is when the direction through the resistor is from negative to positive

"-" Low Potential to "+" High Potential → Rise

Potential Drop – is when the direction through the resistor is from positive to negative

"+" High Potential to "-" Low Potential → Drop

The power supplies

- **Potential Rise** - Going from the negative end (short line) through the power supply to the positive end (long line) is a Potential Rise
- **Potential Drop** - Going from the positive end (long line) through the power supply to the negative end (short line) is a Potential drop.

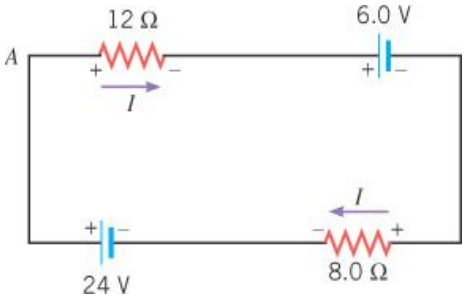
Appling Kirchhoff's Rules

1) Pick the loops of the circuit that are going to be used, just make sure that each element of the circuit is contained in at least one of the loops. Draw the loops on the circuit diagram.
2) Pick a node and draw a direction for the current in each branch or loop of the circuit going into or out of the node. Choose any direction. If your choice is incorrect, the value obtained for the current will turn out to be a negative, but it will not invalidate the results.
3) Mark each resistor with a plus sign at one end and a minus sign at the other end, in a way that is consistent with your choice for the current direction in step 2. Outside the battery,

conventional current is always directed from a higher potential (positive +) toward a lower potential (negative -)

4) Apply the junction rule and the loop rule to the circuit obtaining in the process as many independent equations as there are unknown variables.
5) Solve the equations obtained in step 3 simultaneously for the unknown variables. If a negative current is calculated, the negative must be used in solving other equations.
6) Verify your results. Once you have the values for all currents, you can substitute them back into your original equations to ensure they are correct. Calculating the power supplied versus power dissipated can also serve as a check.

Example: Potential drops and rise around a circuit are determined by the direction you choose to go around the circuit and the rises and drops as defined above. Start anywhere on the circuit and go through each element in the loop in the same direction.



Assume a clockwise rotation and starting from the corner labeled "A".

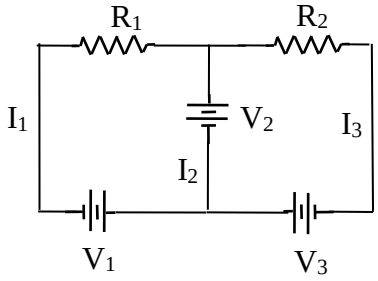
$$\Sigma V_{Drops} = \Sigma V_{Rises}$$

$$[I(12\Omega) + 6\ V + I(8\Omega)]_{Drops} = [24\ V]_{Rises}$$

Solve for I

$$I = \frac{18V}{20\Omega} = 0.9\ A$$

Example: Determine the currents I_1, I_2 and I_3 in the circuit below.

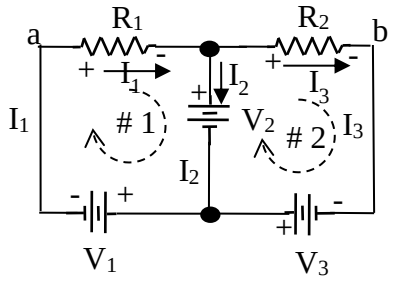


$V_1 = 10\ V$
$V_2 = 20\ V$
$V_3 = 30\ V$
$R_1 = 400\ \Omega$
$R_2 = 500\ \Omega$

- Label the power sources with a "+" next to the long line and "-" next to the short line of the battery.
- Choose a current direction at a node so that there are both incoming currents and outgoing currents.

From the node $I_1 = I_2 + I_3$

Starting at the corner labeled “a” and going in the direction of loop # 1, write down the drops (from + → - as a drop) and rises (from - → + as a rise).



$$(-I_1R_1) + (-V_2) + (-V_1) = 0\ V$$

Solve for I_1

$$I_1 = -\left[\frac{V_2 + V_1}{R_1}\right] = -0.075\ A$$

The chosen direction of I1 was not current it is actually in the opposite direction.

Starting at the corner labeled “b” and going in the direction of loop # 2, write down the drops (from + → - as a drop) and rises (from - → + as a rise).

$$(-I_3R_2) + V_3 + V_2 = 0V$$

Solve for I_3

$$I_3 = \left[\frac{V_3 + V_2}{R_2}\right] = 0.10\ A$$

Using the Junction rule and the current equation

From the node $I_1 = I_2 + I_3$

$$I_2 = I_3 - I_1 = 0.10\ A - (-0.075A) = 0.175\ A$$